

Concept bottleneck models with psychometric attributes

Can a vision model "see" along the same latent axes as humans?

Lepadatu Teodor Pascaru Dan-Alexandru
Students at University of Bucharest

Dan Theodor Oneata Antonio Barbalau
Bitdefender

Introduction

Standard vision models often operate as naive systems. Concept Bottleneck Models (CBM) aim to constrain the predictions so that they align with human-interpretable concepts, therefore offering improved interpretability.

However, the attributes used in CBMs are usually chosen ad-hoc. This research addresses this limitation and investigates a new set of attributes, which are derived from psychometric measurements.

By replacing the ad-hoc attributes with this psychometric foundation, we aim to answer our research question: “Can a vision model see along the same latent axes as humans?”.

Dataset

This work builds on two complementary datasets that together enable the investigation of human-aligned visual concepts: THINGS and SPoSE-49.

THINGS (Hebart et al., 2019) is a large-scale, systematically curated database of 1,854 concrete object categories, each represented by multiple high-quality natural images: animals, vehicles, tools, food, clothing, body parts, household items.

SPoSE-49 (Hebart et al., 2020) is a 49-dimensional, sparse and non-negative embedding derived from approximately 4.7 million human triplet odd-one-out judgements over the THINGS categories providing for each object a vector of carefully selected, human-interpretable attributes that capture semantically meaningful axes of mental representation such as animacy, food-relatedness, mode of transport, colour, material, and bodily association.

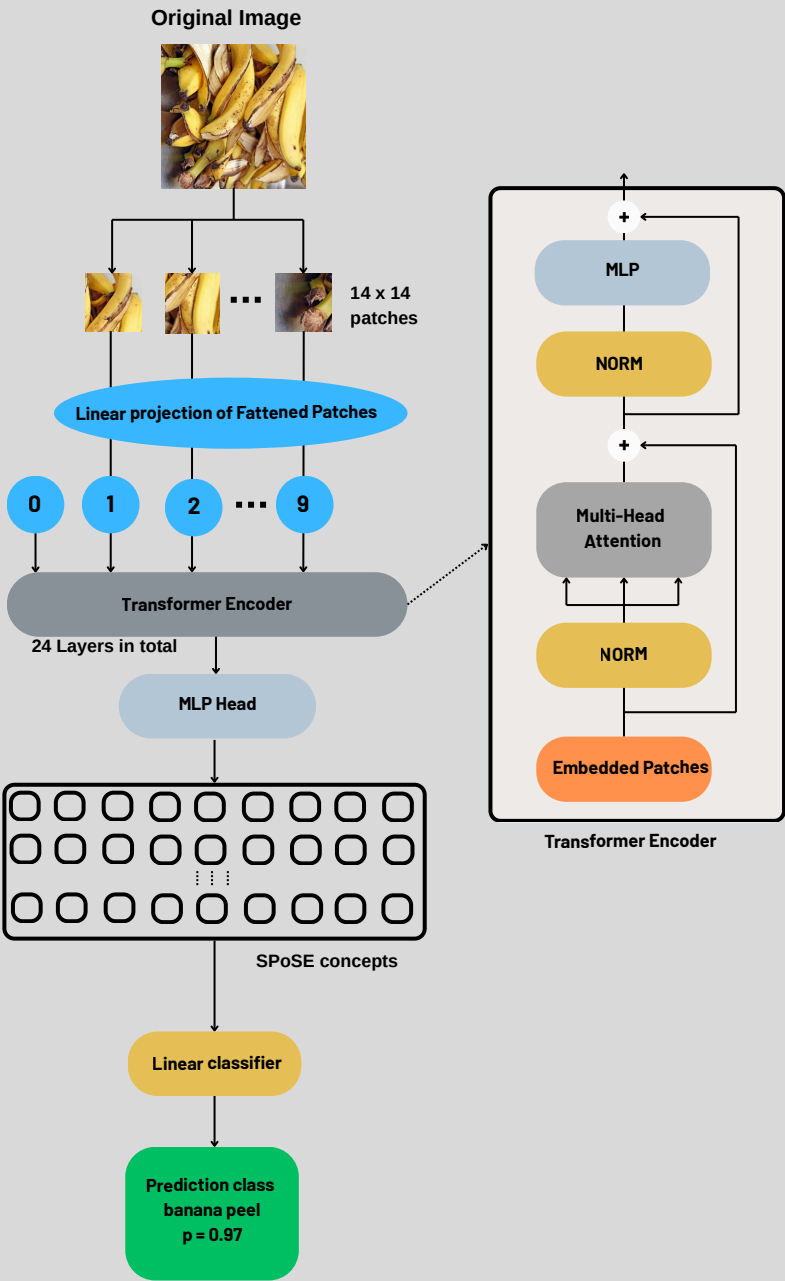
Conclusion and future work

Integrating psychometric attributes into CBMs provides a principled foundation. While these human-derived concepts are theoretically helpful, models struggle to accurately predict them from visual data.

During the predicting phase, the models tend to degrade, or even destroy the integrity of these latent axes. This highlights a massive disconnection between human conceptions and current machine representations.

Future research should focus on exploring alternative architectures and training paradigms capable of bridging the gap between human and machine vision by reliably extracting and preserving psychometric concepts from raw visual data.

Method



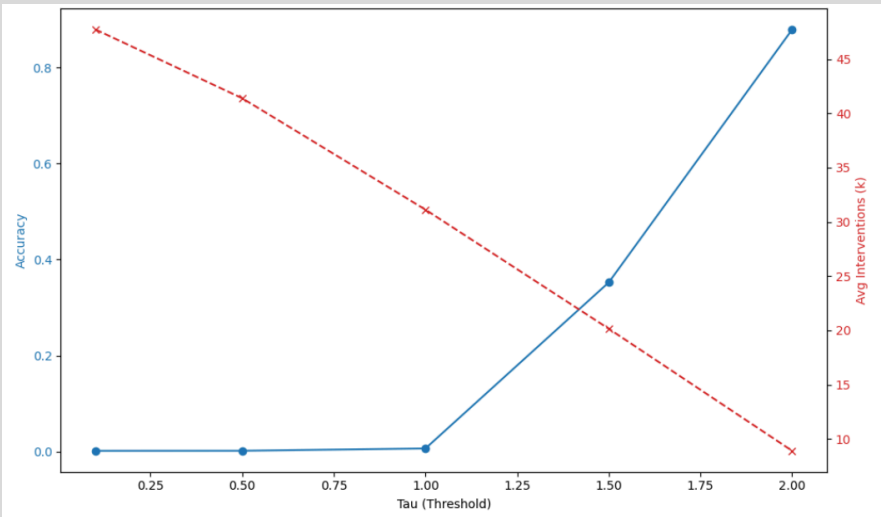
Loss functions:

- Joint: $\text{CrossEntropy}(y, \hat{y}) + \lambda \cdot \text{MSE}(c, \hat{c})$
- Independent and sequential: $\text{CrossEntropy}(y, \hat{y})$, where independent trains f using the true concepts c , while sequential trains f using the predicted concepts $\hat{c} = g(x)$

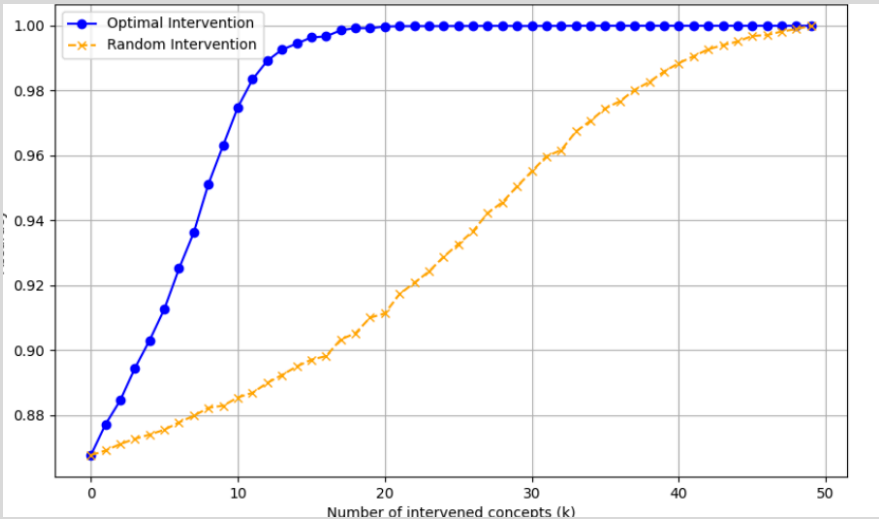
Results

Loss type	Accuracy ↑	F1↑	Precision↑	Recall↑	Loss↓	Mse↓	R ² ↑
Independent	95.9	95.55	96.85	95.7	0.6485	0.0029	0.96
Sequential	96.11	95.78	96.98	95.88	0.9623	0.0029	0.96
Joint	96.5	96.27	97.3	96.37	1.739	2.0352	-35.03

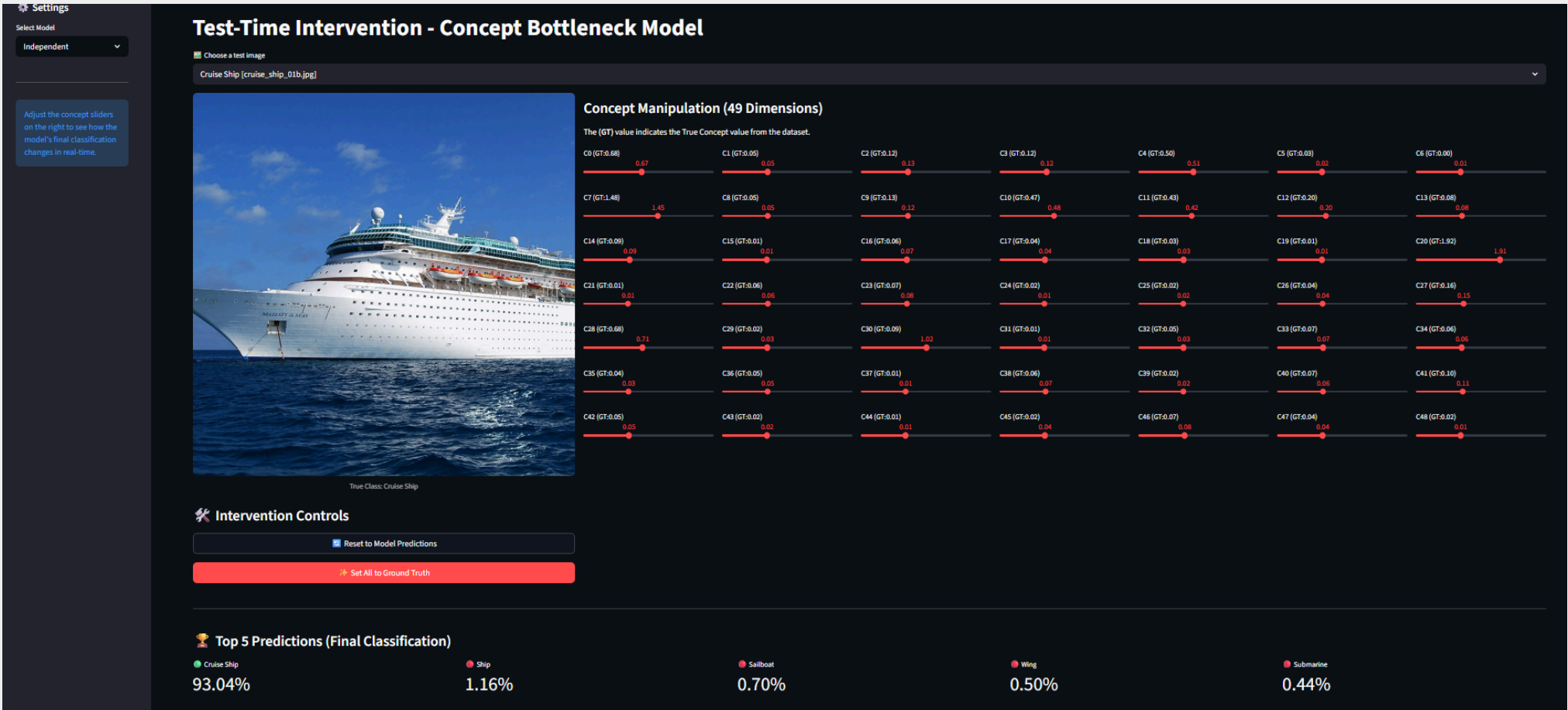
Loss results (MSE and R² are for concept prediction, while the rest for class prediction)



Test time intervention for joint model (lambda = 1.0)



Test time intervention for sequential and independent models



Test time intervention

To evaluate the true utility of the psychometric concepts independently of the vision model's poor extraction capabilities, we utilize test-time interventions.

During inference, we manually override the model's flawed concept predictions by replacing them with the ground-truth SPoSE attribute values.

This isolates the bottleneck layer, allowing us to determine whether correcting these latent semantic axes can successfully recover the downstream classification accuracy that was lost during the initial prediction phase.

We have created a simple web app in which you can choose a model and an image and then adjust the values of the SPoSE concepts. The chosen model is then going to take those modified values and output the top 5 predictions for that image, alongside the probabilities for each.

References:

- Hebart, M.N., Zheng, C.Y., Pereira, F. et al. Revealing the multidimensional mental representations of natural objects underlying human similarity judgements. Nat Hum Behav 4, 1173–1185 (2020).
- Hebart MN, Dickter AH, Kidder A, Kwok WY, Corriveau A, Van Wicklin C, et al. (2019) THINGS: A database of 1,854 object concepts and more than 26,000 naturalistic object images. PLoS ONE 14(10): e0223792.
- Koh, Pang Wei, et al. "Concept bottleneck models." International conference on machine learning. PMLR, 2020.